

PressBangalore

Digital News & Media Platform

Product Specification, Module Flows & Development Scope

Prepared for client review based on demo application, discovery discussions, and media-house best practices.

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Status: Demo delivered — Full production scope + visual UI flows

1. Executive Summary

PressBangalore is a professional digital news and media platform inspired by modern social experiences (Instagram-style reels and stories) but purpose-built for verified journalism, reporter governance, editorial control, and investigative reporting. A working UI demo (web + Android APK) has been developed to showcase core journeys.

Client vision: Users and reporters publish news with images, video, and audio; editorial staff review content using professional Incline/Decline workflows; investigations are tracked with unique IR numbers; location- and language-based feeds serve local audiences; advertisements monetize the platform; users communicate via in-app audio and video calls; published stories offer **Original** and **AI View** tabs (Gemini or GPT); and role-based admin panels govern the entire operation.

Deliverable	Status	Description
UI Demo (Web + Android)	Completed	Full navigable demo with mock data — feed, reels, reporter, admin, IR tracking
Backend API Scaffold	Ready	.NET 8 GraphQL API, PostgreSQL schema, JWT auth — awaiting integration
Production Platform	Planned	Separate admin portal, real uploads, notifications, monetization, compliance

2. Platform Overview

2.1 Three-Portal Architecture

For production, the platform is organized into three dedicated experiences (currently combined in the demo for speed):

- **Public Web & Mobile App** — News feed, reels, search, profile, IR public tracker, consumer registration, premium subscriptions.
- **Reporter Portal** — Registration, credential status, ID card, appointment letter, news submission with photo/video/audio, field recordings, earnings view.
- **Editorial Admin Portal** — Separate secure website for Super Admin, Admin, and Sub Admin with dedicated screens for each operational area (not mixed with consumer UI).

2.2 Target Users

Role	Access	Primary Actions
Consumer (Free)	Public app	Read feed, follow stories, endorse, discuss, share, track IR
Premium User	Public app	Ad-free feed, analytics, and can upload photo/video news — admin must approve before publish
Reporter	Reporter portal	Submit news, upload media, view ID card & appointment letter, check status
Sub Admin	Admin portal (limited)	Post review queue, reporter first-level review, basic moderation
Admin	Admin portal	Reporter approval, ads, teams, IR assignment, dashboard analytics
Super Admin	Admin portal (full)	User management, role assignment, system config, audit, all modules
Investigation Officer (IO)	Admin / IR module	Manage assigned IR cases, submit team reports

3. Professional Media Terminology

Per client direction, consumer-facing and editorial language will reflect a credible newsroom — not casual social media wording. The demo currently uses social terms; production will adopt the mapping below.

Social / Generic Term	PressBangalore Professional Term	Context
Like	Endorse	Reader affirms credibility of a story
Comment	Discuss / Reader Response	Moderated discussion thread
Share	Circulate	Share via link or social channels
Follow	Subscribe / Follow Beat	Follow reporter or topic
Approve	Incline	Editorial acceptance — story meets standards
Disapprove / Reject	Decline	Editorial rejection with reason code
Post	Dispatch / News Item	Published or pending news content
Reels	Briefs / Field Clips	Short-form vertical video news
Stories	Headlines Strip / Live Updates	Ephemeral top stories bar
Report (user)	Flag / Tip-off	Report misleading content

3.1 Editorial Integrity — True News Policy

- All user- and reporter-submitted content enters **Pending Editorial Review** before publication.
- Declined items include a reason code (unverified, defamatory, off-topic, duplicate, poor quality).
- Verified reporter badge displayed only after Incline approval of registration.
- Source attribution field mandatory for field reports; optional document upload for evidence.
- Fact-check status labels: *Verified*, *Under Review*, *Correspondent Report*.
- Correction / update log visible on edited dispatches (media-house standard practice).

4. Module Specifications & Brief Descriptions

4.1 News Feed

Location- and language-filtered stream of published dispatches. Supports text, image galleries, embedded audio clips, and inline video. Headlines strip at top for breaking stories. Advertisement slots (top banner, inline, sponsored label) integrated without disrupting reading flow.

4.2 News Dispatch (Post Creation)

Reporters and authorized users create dispatches with title, body, category (News, Business, Entertainment, Professional), city/beat, language, and media attachments. Submissions go to editorial queue. Admin Inclines to publish or Declines with feedback.

4.3 Briefs (Reels)

Full-screen vertical short videos — field clips, explainers, or professional content. Swipe navigation, endorse/discuss/circulate actions. Separate feed tab. Supports entertainment, business, and professional categories per client vision.

4.4 Reporter Registration & Approval

Multi-step registration: personal details, coverage area (beat), languages, ID document upload. Status lifecycle: *Applied* → *Under Review* → *Inclined (Approved)* / *Declined*. Email/SMS notification at each stage. Only Inclined reporters can submit dispatches.

4.5 Reporter Credentials

Digital press ID card with photo, name, beat, validity, QR verification. Appointment letter for assigned area — viewable and downloadable as PDF. Reporter status dashboard: active, suspended, pending renewal.

4.6 Audio & Video Attachments

News dispatches support image, video (MP4), and audio (MP3/WAV) attachments. In-app field recorder for reporters (video/audio capture). Server-side transcoding and thumbnail generation in production. Max file size and duration policies configurable.

4.7 User & Premium User

Free consumers: read, endorse, discuss, circulate. Premium tier: ad-free feed, priority support, profile analytics, and **can upload photo/video news** from the app. All Premium submissions go to the editorial queue — admin must Incline before publish; Declined items return to the user with a reason. Subscription billing in a later release.

4.8 Editorial Approval Flow

Central workflow engine: every dispatch and reporter application passes through review queues. Sub Admin may perform first review; Admin/Super Admin final Incline or Decline. Full audit trail: reviewer, timestamp, action, reason. Bulk actions for high-volume days.

4.9 Advertisement Panel

Admin creates ad campaigns: image/video creative, target city, placement slot (Feed Top, Feed Inline, Briefs interstitial), schedule, active toggle. Future: impression/click analytics, advertiser self-serve portal, billing.

4.10 Investigation (IR) Module

Create Investigation Record with auto-generated trackable number (e.g. IR-BLR-2026-00042). Assign Investigation Officer (IO) and field teams. Teams submit consolidated reports. Public tracker page: enter IR number, view status timeline and merged findings.

4.11 Teams & Report Consolidation

Define teams (name, members, specialization). Assign teams to IR cases. Each team submits field reports; IO consolidates into master investigation summary. Status stages: Open → Assigned → In Progress → Under Review → Closed.

4.12 User Profile & Dashboard

Profile: avatar, bio, dispatches grid, endorsements received, subscriber count. Dashboard: views over time, top dispatches, earnings (reporters/premium creators), engagement breakdown. Charts and export for admin analytics.

4.13 Search & Discovery

Full-text search across dispatches, reporters, tags. Trending topics for city. Quick link to IR public tracker. Filter by language, category, date.

4.14 In-App Communication (Audio & Video Calls)

Users, reporters, and editorial staff can communicate directly within the platform. One-to-one and group audio calls; one-to-one and group video calls; call from profile, dispatch thread, or reporter directory. Incoming call UI, mute, camera toggle, and call history. Reporters can reach editors; teams on an IR case can join a group call. Built with WebRTC (browser/mobile) and SignalR signaling on the existing .NET API.

4.15 AI News View (Original & AI Tabs)

Design choice (recommended): inside the article page — not a separate outside page. When a user opens any published dispatch, two tabs appear at the top: **Original** (reporter's exact text and media) and **AI View** (AI-generated summary, simplified language, key bullet points, optional Kannada/English/Hindi rewrite). User taps to switch — same story, same page, no extra navigation. When admin inclines a dispatch to publish, the backend calls **Google Gemini** or **OpenAI GPT** (API key configured in admin settings, stored server-side only) to generate and save the AI version alongside the original. AI output is clearly labelled. Admin can toggle AI on/off, pick provider, and regenerate from admin portal.

5. Editorial Admin Portal — Separate Screen Structure

Client requirement: dedicated admin website with isolated screens per function (not embedded in consumer app). Each screen is role-gated.

Admin Screen	Purpose	Roles
Dashboard Overview	KPIs: users, pending reporters, pending dispatches, earnings, IR count	All admins
Dispatch Review Queue	Incline / Decline submitted news with media preview	Sub Admin+
Reporter Applications	Review credentials, Incline or Decline registration	Admin+
Reporter Directory	Active reporters, status, beat, suspend/reinstate	Admin+
User Management	Consumers, premium status, role assignment	Super Admin
Advertisement Manager	CRUD ads, slots, scheduling, city targeting	Admin+
Investigation (IR) Center	Create IR, assign IO, assign teams, view timeline	Admin+
Team Management	Create teams, members, assignment history	Admin+
Report Consolidation	Merge team field reports per IR case	Admin, IO
Analytics & Earnings	Charts, export, reporter payouts summary	Admin+
System Settings	Cities, languages, categories, reason codes, notifications	Super Admin
AI Settings	Gemini/GPT provider, API key, enable AI View, regenerate	Super Admin
Audit Log	Immutable log of all editorial and admin actions	Super Admin

6. Visual User Flow Diagrams

Below is a **simple summary table** of all flows, then **screen-by-screen pictures** for each. Phone frames = public app. Wider panels = admin website. Green = **Incline** (approve). Red = **Decline** (reject).

6A. All Flows at a Glance (One Line Each)

Flow	Name	One-Line Explanation (Simple)
1	Consumer Journey (Read & Engage)	A normal person opens the app, picks city and language, reads and watches local news, and can share opinions — Premium members can also submit news (see Flow 3).
2	Reporter Journey (Register → Publish)	A journalist registers, admin approves, they get a press ID card, then every news they post is checked by admin before it goes public.
3	Premium User Posts News (Upload & Approval)	A Premium member uploads photo or video news from their phone; admin must approve (Incline) before it appears in the feed for everyone.
4	Editorial Approval (True News Check)	All news — from reporters or Premium users — waits in a queue until admin clicks Incline (publish) or Decline (reject); nothing goes live without this.
5	Investigation Case (IR Tracking)	Admin opens an investigation with a unique case number; teams send reports; anyone can type that number in the app and see how the case is progressing.
6	Advertisements & Premium Membership	Admin puts ads in the news feed; users can pay for Premium (extra benefits); reporters and creators can see how much they earned.
7	Audio & Video Calls (Talk Inside App)	Users, reporters, and editors can make audio or video calls inside the app to discuss news or work together — no need to switch to WhatsApp.
8	AI News View (Original vs AI Tabs)	When reading a published story, the user switches between Original (reporter text) and AI View (simple summary / translation) using two tabs on the same article page.

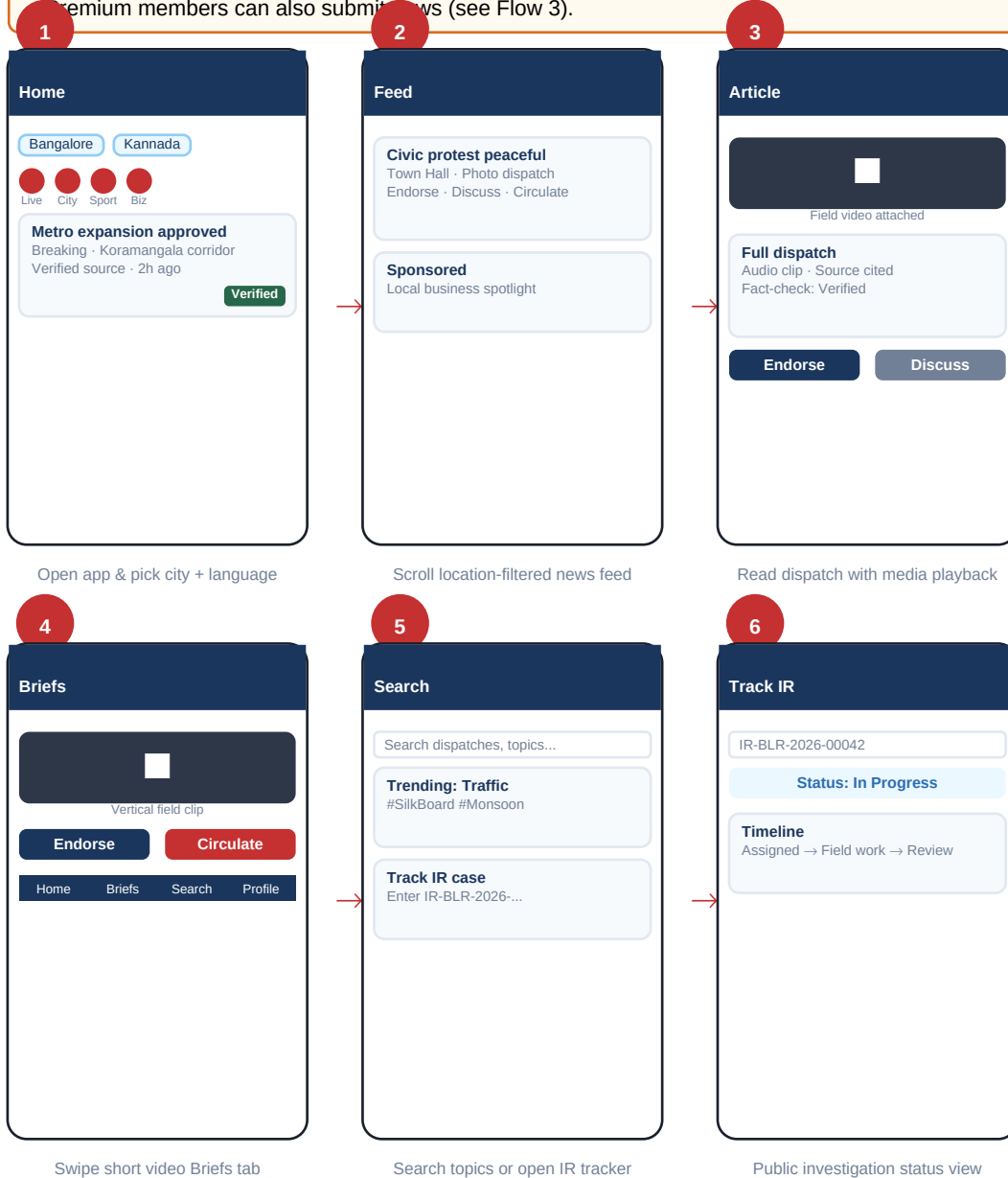
6B. Visual Screen Flows (Step by Step)

Flow 1 — Consumer Journey (Read & Engage)

Public app · Read news · Watch Briefs · Search · Track cases

In simple words:

A normal person opens the app, picks city and language, reads and watches local news, and can share opinions — premium members can also submit news (see Flow 3).

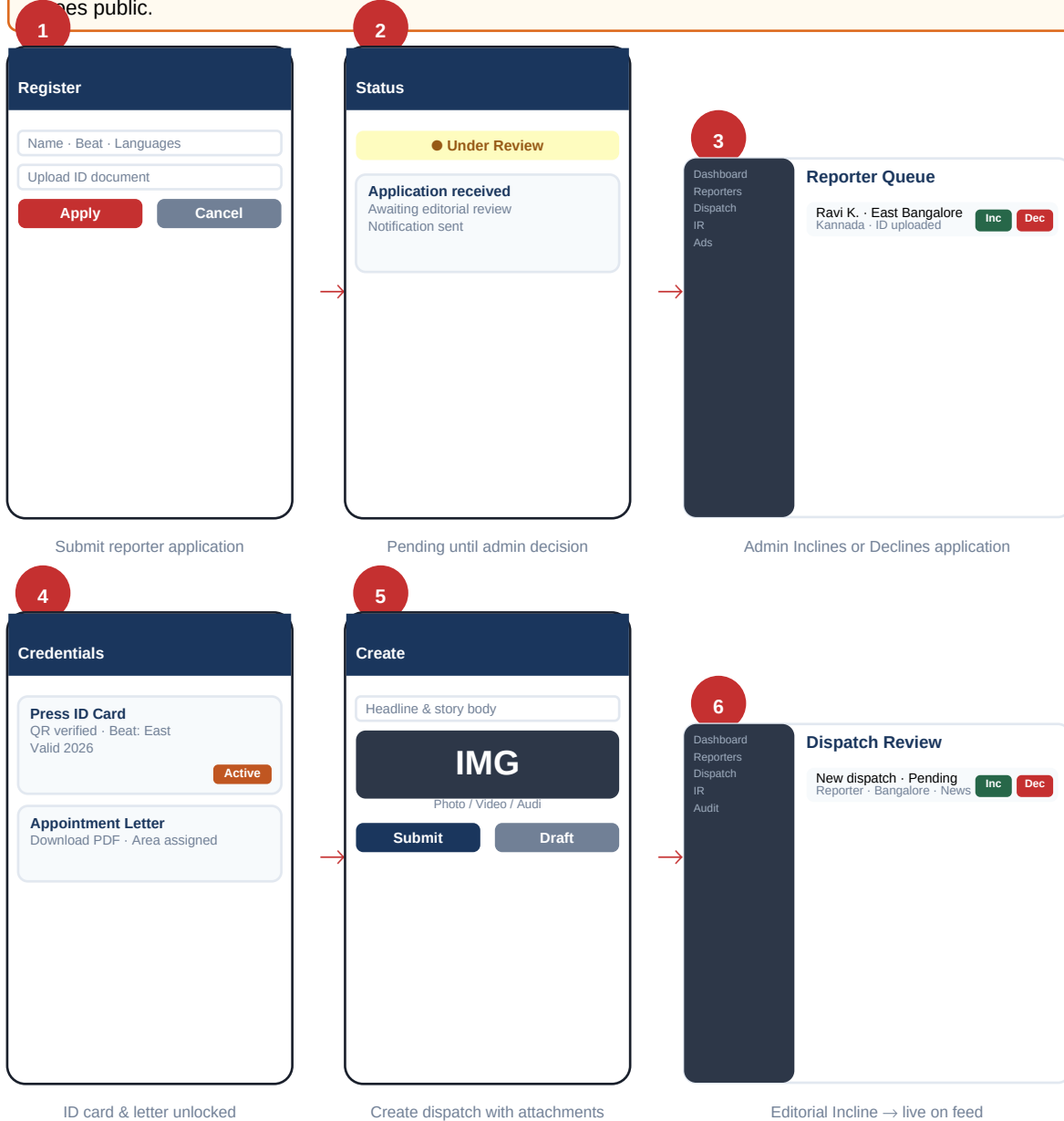


Flow 2 — Reporter Journey (Register → Publish)

Apply · Admin approval · Press ID · Post news · Admin checks again

In simple words:

A journalist registers, admin approves, they get a press ID card, then every news they post is checked by admin before it goes public.

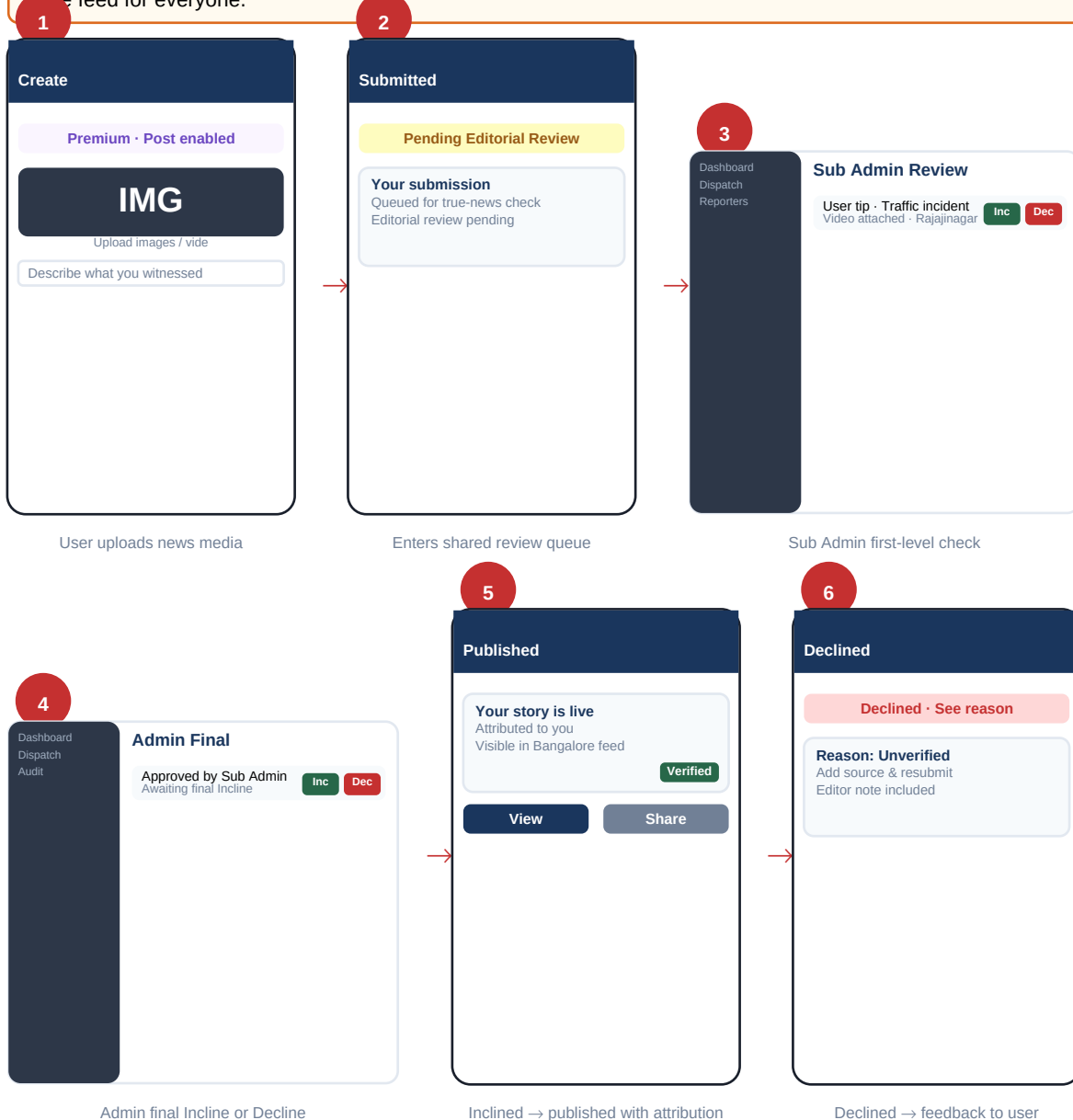


Flow 3 — Premium User Posts News (Upload & Approval)

Upload photo/video · Admin checks · Incline or Decline · No auto-publish

In simple words:

A Premium member uploads photo or video news from their phone; admin must approve (Incline) before it appears in the feed for everyone.

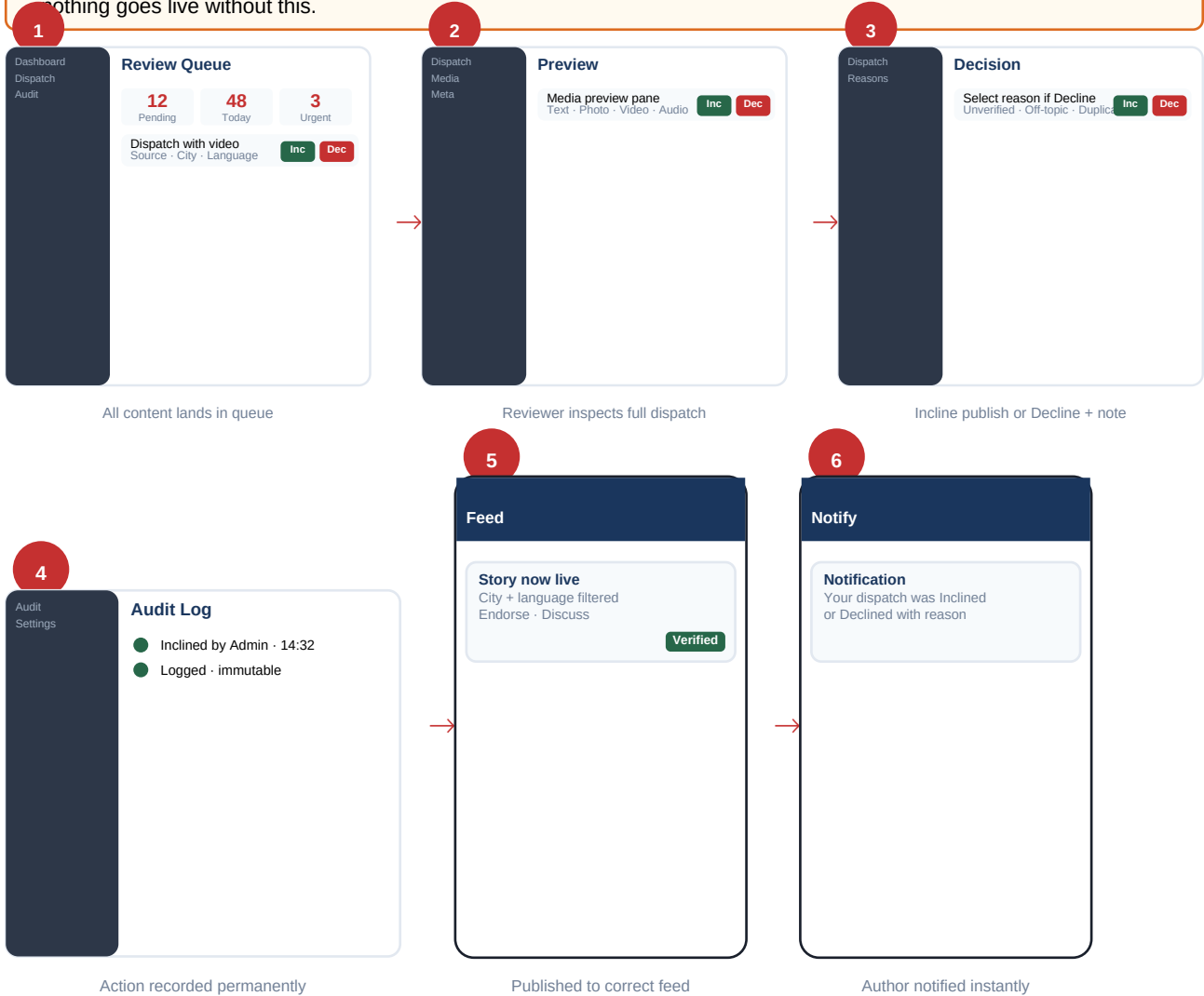


Flow 4 — Editorial Approval (True News Check)

Review queue · Preview story · Incline or Decline · Audit log

In simple words:

All news — from reporters or Premium users — waits in a queue until admin clicks Incline (publish) or Decline (reject); nothing goes live without this.

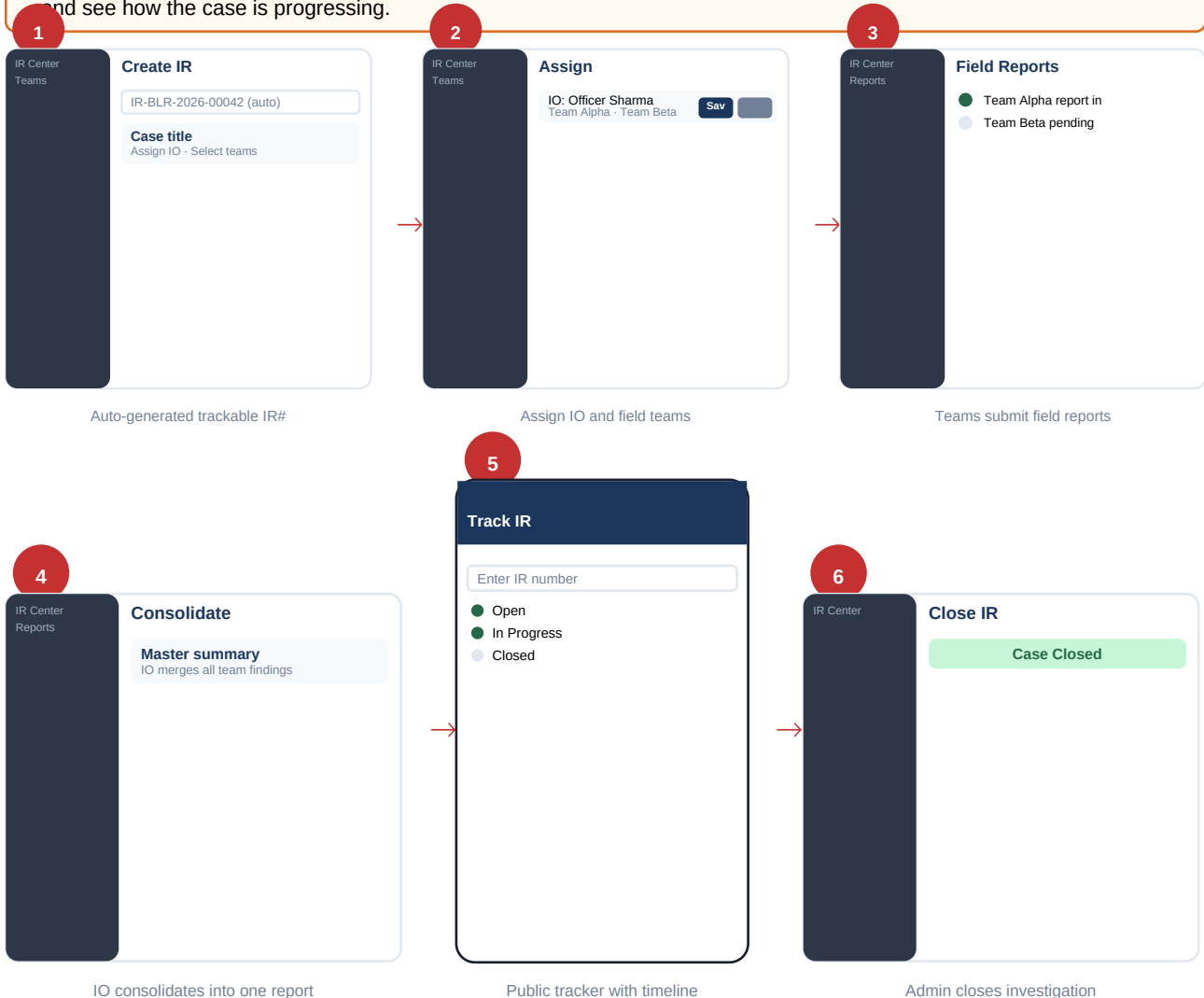


Flow 5 — Investigation Case (IR Tracking)

Case number · Assign teams · Field reports · Public status check

In simple words:

Admin opens an investigation with a unique case number; teams send reports; anyone can type that number in the app and see how the case is progressing.

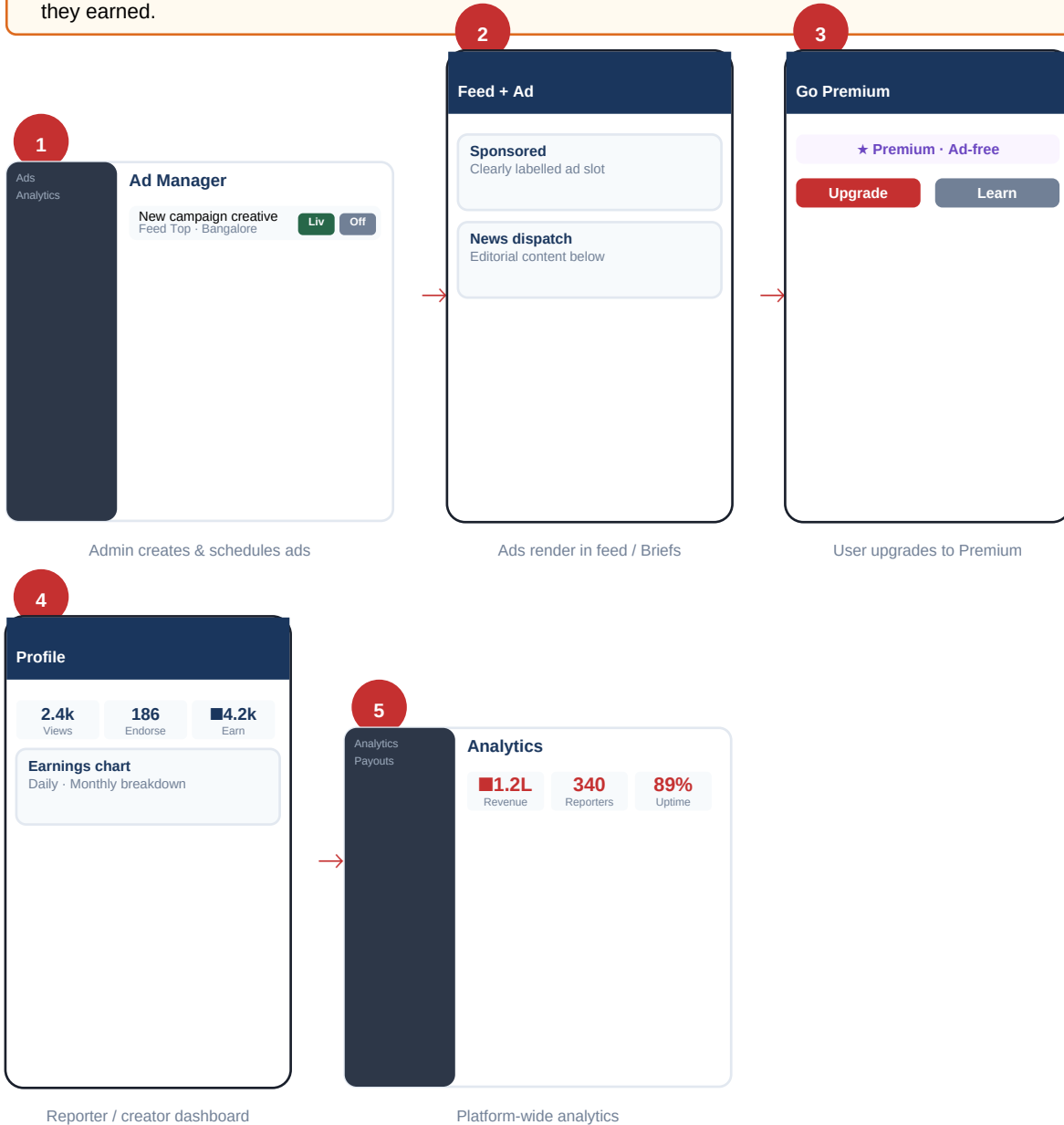


Flow 6 — Advertisements & Premium Membership

Ads in feed · Premium upgrade · Earnings dashboard

In simple words:

Admin puts ads in the news feed; users can pay for Premium (extra benefits); reporters and creators can see how much they earned.

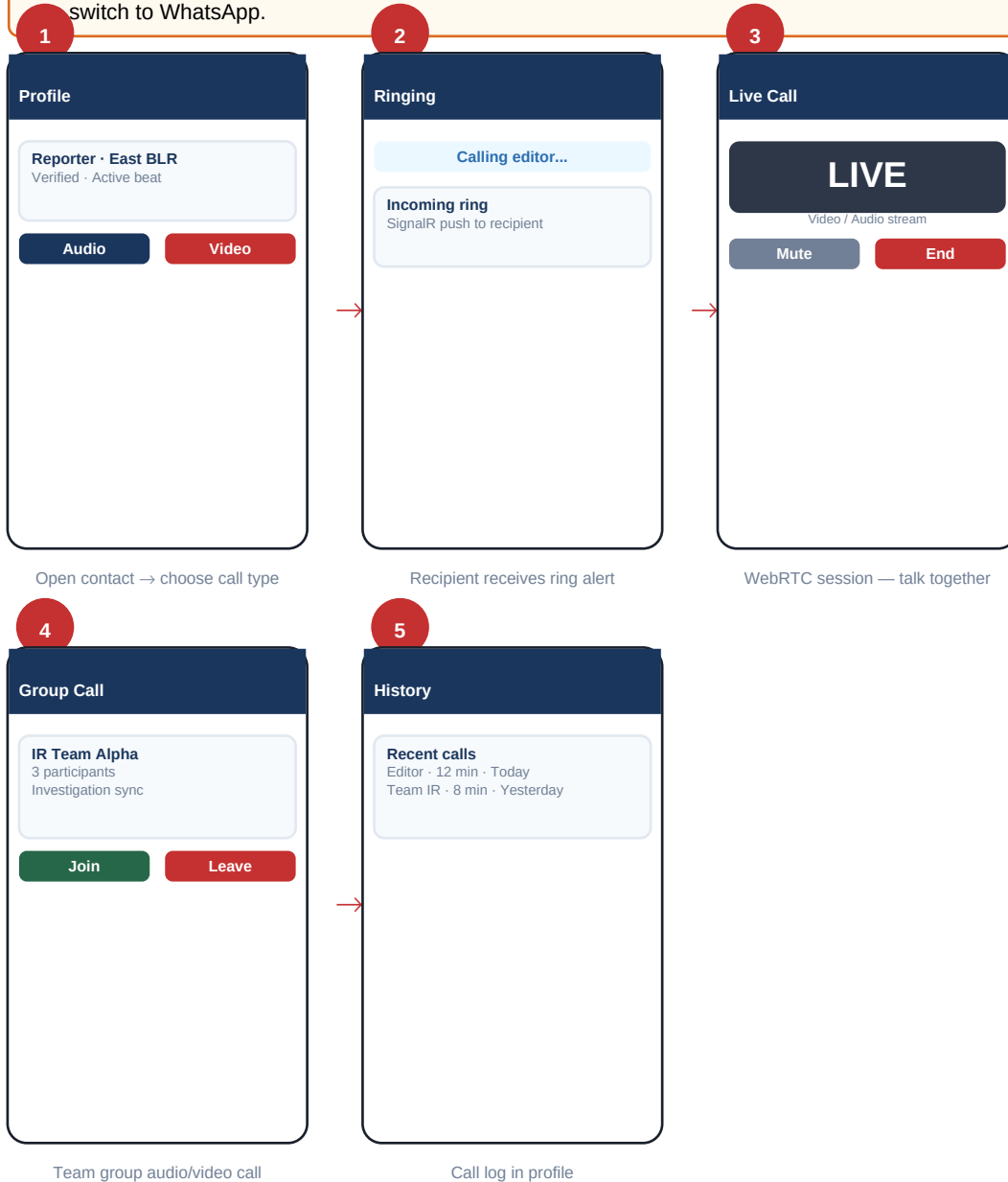


Flow 7 — Audio & Video Calls (Talk Inside App)

Call button · Ring · Live talk · Group calls · Call history

In simple words:

Users, reporters, and editors can make audio or video calls inside the app to discuss news or work together — no need switch to WhatsApp.

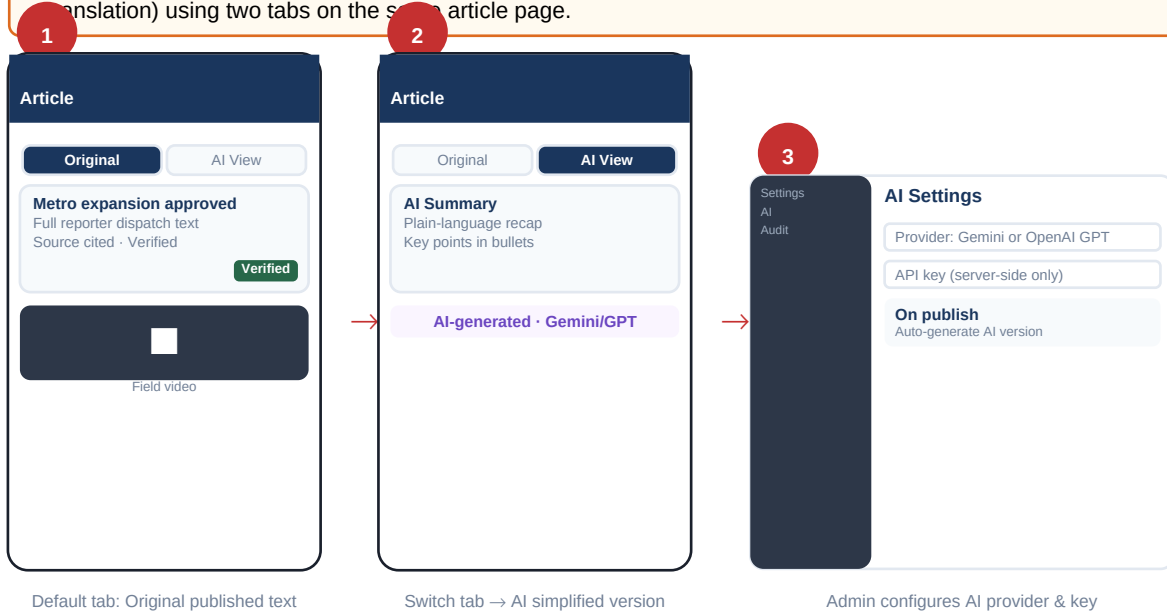


Flow 8 — AI News View (Original vs AI Tabs)

Inside article page · Two tabs · Gemini or GPT · Generated on publish

In simple words:

When reading a published story, the user switches between Original (reporter text) and AI View (simple summary / translation) using two tabs on the same article page.



6C. Flow Step Details (Quick Reference)

Short step list for each flow — for team reference during development and client walkthrough.

Flow 1 — Consumer (Read News)

In one line: A normal person opens the app, picks city and language, reads and watches local news.

1. Open app → select city (e.g. Bangalore) and language (e.g. Kannada).
2. Scroll the news feed — stories, photos, videos, and ads.
3. Tap a story → read full article; switch **Original** or **AI View** tabs (see Flow 8).
4. Endorse (trust it), Discuss (comment), or Circulate (share).
5. Open Briefs tab → swipe short news videos.
6. Search any topic, or track an investigation case number.
7. Note: Premium members can also post news — see Flow 3.

Flow 2 — Reporter (Journalist)

In one line: A journalist registers, gets approved, receives press ID, posts news after admin check.

1. Apply as reporter — name, area, languages, ID document.
2. Wait — status shows Under Review.
3. Admin Inclines (approves) or Declines (rejects) in admin portal.
4. On approval → digital press ID card and appointment letter.
5. Create news with photo, video, or audio attached.
6. Admin Inclines dispatch → published to city/language feed.

Flow 3 — Premium User Posts News

In one line: Premium member uploads photo/video news; admin approves before it goes public.

1. Premium user taps Create / Submit news.
2. Upload images or video + write what happened.
3. Status: Pending Editorial Review — not live yet.
4. Sub Admin does first check.
5. Admin gives final Incline or Decline.
6. Incline → story appears in feed with user's name.
7. Decline → user gets reason and can fix and resubmit.

Flow 4 — Editorial Approval (All News)

In one line: Every story waits for admin Incline or Decline — true news policy.

1. Reporter or Premium user submission enters review queue.
2. Admin previews text, photos, videos, and details.
3. Click Incline to publish or Decline with reason.
4. Action saved in audit log permanently.

5. Author gets notification; published story appears in feed.

Flow 5 — Investigation Case (IR)

In one line: Admin opens a case with tracking number; public can check status.

1. Admin creates case — system gives number (e.g. IR-BLR-2026-00042).
2. Assign investigation officer and field teams.
3. Teams submit field reports; officer merges them.
4. Public user enters number on Track page → sees timeline.
5. Admin closes case when done.

Flow 6 — Ads & Premium

In one line: Admin runs ads; users upgrade to Premium; reporters see earnings.

1. Admin creates ad — image/video, city, placement slot.
2. Ads show in feed and Briefs (clearly labelled).
3. User upgrades to Premium for ad-free and extra features.
4. Reporter views earnings — views, endorsements, income.
5. Admin sees platform-wide analytics and payouts.

Flow 7 — Audio & Video Calls

In one line: People talk inside the app — audio or video — no WhatsApp needed.

1. Open a profile → tap Audio Call or Video Call.
2. Other person gets ring notification.
3. They accept → live call starts.
4. Mute, turn camera off, or hang up during call.
5. IR teams can do group calls for case coordination.

Flow 8 — AI News View (Original vs AI)

In one line: On the same article page, user switches between real story and AI simple version.

1. User opens a published news story (inside the app).
2. Two tabs at top: Original | AI View — on the same page (not separate website).
3. Original tab → reporter's exact text, photos, and video.
4. AI View tab → simple summary, bullet points, easy language (or translation).
5. Label shown: AI-generated via Gemini or GPT.
6. When admin publishes news, server auto-creates AI version using API key.
7. Admin sets Gemini or GPT key in AI Settings (key never shown to users).

7. Recommended Additions (Media House Best Practices)

Based on industry standards for digital newsrooms, the following enhancements are recommended beyond the stated client requirements:

Enhancement	Benefit
Breaking News Push Alerts	Editor-triggered push notifications for verified breaking dispatches by city.
Editor's Pick / Featured Slot	Curated top story placement on feed — admin-controlled.
Content Versioning	Track edits after publication; show 'Updated' timestamp and change summary.
Tip-off / Whistleblower Channel	Anonymous tip submission with encrypted handling for investigative desk.
Press Release Portal	Separate channel for official government/corporate statements.
Multi-language CMS	Admin UI and dispatch content in Kannada, English, Hindi with translation workflow.
Reporter Beat Map	Visual map of coverage areas; assign exclusive beats to avoid duplicate reporting.
Engagement Analytics for Editors	Heatmaps: which topics, cities, and formats perform best.
GDPR / Data Privacy Module	Consent management, data export, account deletion — essential for user trust.
Two-Factor Auth for Admins	Mandatory 2FA for all editorial admin accounts.
Watermark on Field Media	Automatic PressBangalore watermark on reporter uploads for attribution.
Live Broadcast	Live video stream for press conferences and breaking events.
RSS & Syndication API	Allow partner sites to syndicate published feeds with attribution.
AI Original / AI View Tabs	Inside article page — switch between reporter text and AI summary (Gemini/GPT).
Print-Ready Export	Generate PDF of dispatch for print edition compatibility.

8. Technology Stack & Infrastructure

The chosen stack is suitable for the full platform including news, admin workflows, and real-time audio/video communication. **Render** hosts the API and services; **Cloudflare** handles CDN, DNS, SSL, and media edge delivery — together this is enough for production launch at the proposed scale.

Layer	Technology	Notes
Consumer & Reporter UI	React 19, TypeScript, Vite, Tailwind CSS	Responsive web + Capacitor Android app
Admin Portal	React (separate deployment)	Dedicated subdomain e.g. admin.pressbangalore.com
API	.NET 8, HotChocolate GraphQL	Queries/mutations for all domain operations
Real-time / Calls	SignalR + WebRTC	Call signaling on API; peer audio/video in app
AI Layer	Google Gemini or OpenAI GPT	API key on server; generates AI View on publish
File Upload	REST multipart + Cloudflare R2/CDN	Images, video, audio attachments
Database	PostgreSQL (managed)	Full relational schema with audit tables
Auth	JWT + role-based access control	BCrypt passwords; 2FA for admins (recommended)
Hosting — API	Render	.NET API service, SignalR WebSockets, background jobs
Hosting — Frontend	Cloudflare Pages / CDN	Static web app, fast global delivery
DNS & Security	Cloudflare	SSL, DDoS protection, caching, optional R2 storage
Mobile	Capacitor 8 (Android)	Same codebase as web; APK + installable PWA

8.1 Audio & Video Calls — Technology Fit

- **Signaling** — ASP.NET Core SignalR on Render manages call setup, ring, accept, decline, and hang-up.
- **Media** — WebRTC in React (web) and Capacitor (Android) handles live audio and video streams peer-to-peer.
- **STUN/TURN** — Cloudflare (or a lightweight TURN add-on) ensures calls work across mobile networks and firewalls.
- **Render + Cloudflare is sufficient** — no separate heavy media server required for standard 1-to-1 and small group calls.
- Call features integrate with existing user roles: consumers, reporters, editors, and IR teams can call permitted contacts.

8.2 AI News View — Design & Technology

Where to put it: Inside the app, on the **same article detail page** — not a separate outside website. Two tabs (*Original* | *AI View*) sit below the headline. This is the best UX: user stays on one story, taps to compare real vs simplified version.

- **Original tab** — exact published dispatch from reporter or Premium user (text, images, video).
- **AI View tab** — generated summary, bullet key points, plain language, optional language rewrite.
- **Provider** — Google Gemini API or OpenAI GPT API; admin chooses one in AI Settings.
- **API key** — stored only on Render server (environment variable); never exposed to mobile/browser.
- **When generated** — automatically when admin Inclines (publishes) a dispatch; cached in database.
- **Label** — AI View always shows “AI-generated” badge so readers know it is not the original report.
- **Admin controls** — enable/disable AI, switch provider, regenerate AI text for a story.

9. Development Phases & Deliverables

Phase 0 — Demo (Completed)

UI demo with mock data; Android APK; all hero journeys navigable; client presentation ready.

Phase 1 — Production Foundation

Backend integration; real auth and file upload; separate admin portal; Incline/Decline workflow; reporter registration and dispatch queues; news feed with filters; IR module; basic ads; audio/video call signaling and 1-to-1 calls; AI View (Gemini/GPT) on publish.

Phase 2 — Growth & Monetization

Premium subscriptions; payment gateway; push notifications; PDF ID cards and appointment letters; earnings dashboard; analytics; email/SMS notifications; group audio/video calls.

Phase 3 — Advanced Media

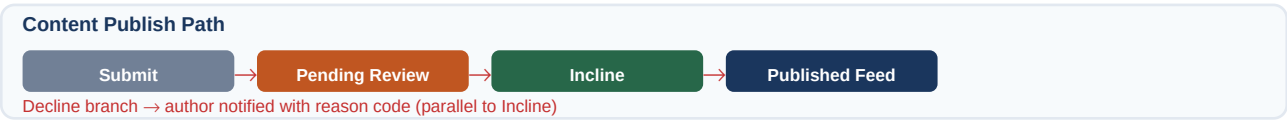
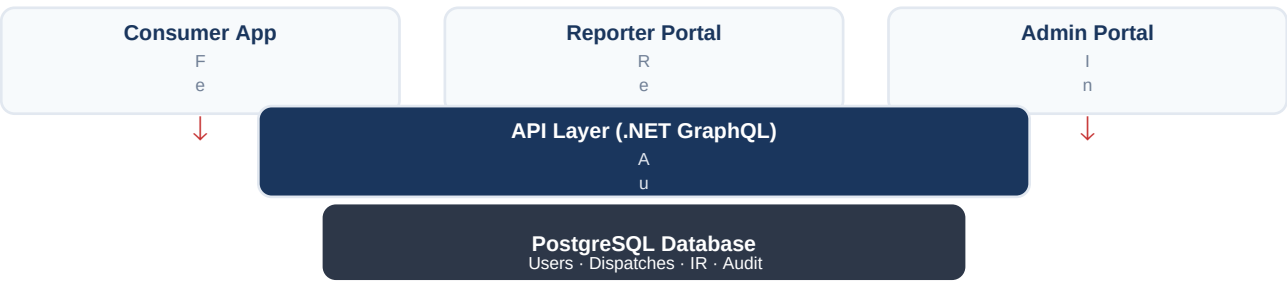
Live broadcast; advertiser self-serve; syndication API; multi-language CMS; tip-off channel; advanced ad targeting and billing.

9.1 What Will Be Developed — Summary Checklist

- Public news feed (location + language filters)
- News dispatch with image, video, audio attachments
- Briefs (reels) — short vertical video feed
- Reporter registration, Incline/Decline approval, status tracking
- Digital reporter ID card and appointment letter (PDF)
- In-app audio/video field recording for reporters
- User and Premium user accounts with profile dashboard
- Premium user upload news (photo/video) with editorial approval
- Editorial Incline/Decline approval for all submitted content
- Separate admin portal with dedicated screens per function
- Super Admin / Admin / Sub Admin role hierarchy
- Advertisement panel with feed and Briefs placements
- IR investigations with trackable number and public tracker
- IO assignment and multi-team report consolidation
- Professional terminology (Endorse, Discuss, Incline, Decline)
- True news policy with source attribution and audit trail
- Search, analytics, earnings information
- In-app audio and video calls (user-to-user, reporter-to-editor, team calls)
- AI View tab on article page (Original vs AI summary — Gemini or GPT)
- Web application + Android mobile app (single codebase)

10. High-Level System Architecture

Three portals connect through a single API layer to PostgreSQL. All submitted content follows the publish path at the bottom.



11. Conclusion

The PressBangalore demo validates the user experience and core journeys discussed with the client. This document defines the full production scope: a three-portal architecture with professional editorial language, mandatory content review for true news integrity, separate admin screens, in-app audio and video communication, and comprehensive modules for reporters, investigations, advertisements, and analytics. The React + .NET + PostgreSQL stack on Render and Cloudflare is well suited for web, mobile, and real-time calls.

12. Project Budget

Item	Details
Total project cost	■3,00,000 (Rupees Three Lakhs Only)
Deliverables included	Public web application · Android mobile app · Reporter portal · Editorial admin portal · Backend API · News feed, Briefs, approvals, IR module, advertisements, analytics, audio & video calls, AI Original/AI View tabs

Item	Details
Hosting (client / operational)	Render (API + services) · Cloudflare (CDN, DNS, SSL, media edge) — sufficient for launch; hosting costs billed separately to client infra accounts
Demo status	UI demo completed — production build per scope above

Note: Budget covers design, development, testing, and deployment setup for web and mobile as described in this document. Third-party paid services (SMS gateway, payment gateway, domain registration, Render/Cloudflare usage beyond free tiers, and Gemini/GPT API usage charges) are excluded unless agreed separately.

Document prepared by Haja Mainudeen, Full Stack Developer (7 years experience). For questions or scope adjustments, please contact directly.